

Do Attention Weights Reflect Token Importance?

A Small-Scale Study with DistilBERT on Sentiment Classification

Anonymous DLCC submission

Paper ID 0001

Abstract

Attention weights are often used as a convenient explanation of which words matter most in a Transformer prediction, but prior work has questioned whether attention reliably reflects token importance. We revisit this question in a controlled setting by fine-tuning DistilBERT on the SST-2 sentiment classification task and repeating the experiment across four random seeds. For each validation example, we rank tokens using three signals: last-layer [CLS] attention, gradient $\times$ input, and leave-one-out (LOO) deletion. We compare these rankings with Spearman correlation and evaluate them with token-deletion tests on the full validation set, measuring confidence drops and prediction flips after removing top-ranked tokens. Across seeds, the classifier achieves stable validation performance, with mean accuracy 0.9074 ± 0.0052 . However, agreement between ranking methods is weak: attention has mean Spearman correlation 0.224 with gradient $\times$ input and 0.176 with LOO. Deletion experiments show a clearer hierarchy. Removing the top LOO-ranked token yields the largest mean confidence drop (0.0383) and highest flip rate (0.302), while attention-based deletion is stronger than random deletion but substantially weaker than LOO. In our setting, attention-based deletion also exceeds gradient $\times$ input deletion, suggesting that raw attention contains some useful signal even though it does not align closely with alternative importance definitions. Overall, our findings suggest that attention is informative but not a reliable standalone proxy for token importance. The project provides a compact and reproducible empirical benchmark for testing interpretability claims in Transformer-based text classification.

1. Introduction

Transformer models are built around attention and have become a dominant architecture for natural language processing since the introduction of the Transformer itself [10], BERT [3], and compact variants such as DistilBERT [6].

Because attention weights are produced internally by the model and can be visualized directly, they are often treated as a natural explanation of which input tokens matter most for a prediction. In sentiment classification, for example, one might expect tokens with higher attention to correspond to the words that most strongly support the predicted label.

That intuition is appealing, but it remains controversial. Prior work has shown that attention weights do not necessarily align with other feature-importance measures or faithfully reflect how much a token influences the output [4, 7]. At the same time, later discussion argues that the usefulness of attention depends on what one expects from an explanation and how the comparison is performed [12]. This debate makes attention a good target for a focused empirical study: if attention is to be used as an explanation, it should at least show consistent agreement with alternative importance signals or reliably identify tokens whose removal changes the model's decision.

In this paper, we ask a narrow question: *do attention weights reflect token importance in a Transformer sentiment classifier?* We study this question with DistilBERT fine-tuned on SST-2, a standard binary sentiment task from GLUE [8, 11]. Our analysis compares three token-ranking signals on the full validation split: last-layer [CLS] attention, gradient $\times$ input, and leave-one-out (LOO) deletion. We evaluate them in two complementary ways. First, we measure agreement between rankings using Spearman correlation. Second, we test how disruptive each ranking is by deleting top-ranked tokens and measuring confidence drops and prediction flips.

The paper makes three contributions. First, it provides a compact multi-seed benchmark for comparing attention with gradient-based and perturbation-based token importance signals in a Transformer classifier. Second, it separates two questions that are often conflated: whether methods rank tokens similarly, and whether they identify tokens whose removal most affects the model. Third, it packages the study in a reproducible workflow that produces saved metrics, token rankings, deletion summaries, and paper figures from the same experiment pipeline.

2. Related Work

The interpretability of attention has been debated extensively in NLP. Jain and Wallace [4] argue that standard attention weights often have weak correlation with gradient-based importance measures and that substantially different attention distributions can lead to similar predictions. Serrano and Smith [7] similarly show that large attention weights do not always correspond to large causal effects on model outputs. In response, Wiegrefe and Pinter [12] argue that whether attention counts as an explanation depends on the definition of explanation and on how the evaluation is framed. Our work does not aim to settle this broader debate; instead, it tests a concrete and measurable version of the question in a simple Transformer classification setting.

We compare attention against two established families of attribution methods. Gradient-based methods estimate importance by differentiating the model score with respect to the input, and input $\times$ gradient is one of the best-known simple variants [1]. Perturbation-based methods estimate importance by observing how the prediction changes when part of the input is removed or altered. Representation erasure and leave-one-out style analyses have been used as a direct way to measure contribution by deletion [5]. These two families provide useful comparison points because they capture different notions of importance: local sensitivity around the input versus explicit predictive change under removal.

Our setting is also related to analyses of attention patterns in pretrained Transformers. Clark et al. [2] show that BERT attention often exhibits strong structural patterns, including frequent focus on delimiter tokens and position-based behaviors. Sun and Marasović [9] further argue that raw attention can differ from more output-relevant effective attention. These observations motivate our decision to examine a simple last-layer attention signal empirically rather than assume that visual salience is equivalent to functional importance.

3. Methods and Experimental Setup

3.1. Task, Model, and Training Setup

We study binary sentiment classification on the SST-2 task from GLUE [8, 11]. The base model is `distilbert-base-uncased`, a compact distilled version of BERT [6, 3]. For each seed in {42, 43, 44, 45}, we fine-tune the model for 2 epochs with learning rate 2×10^{-5} , weight decay 0.01, maximum sequence length 128, training batch size 16, and evaluation batch size 32. We select the best checkpoint for each seed using validation accuracy. The analysis uses the full SST-2 validation split of 872 examples rather than a small subsample.

The implementation is organized as a notebook-driven interface over a modular Python pipeline. Training, evalua-

tion, and interpretability analysis are handled by reusable functions, while saved artifacts include validation metrics, token rankings, ranking-similarity summaries, deletion records, and qualitative cases. Figure 1 summarizes the workflow.

3.2. Token-Importance Signals

For an input sequence x with predicted label $\hat{y}$, we assign an importance score to each non-special token i using three methods.

Attention. We use the final Transformer layer and average attention over heads. Let $A^{(L,h)}$ be the attention matrix of head h in the last layer. We score token i using the attention weight from the [CLS] query position:

$$s_i^{\text{attn}} = \frac{1}{H} \sum_{h=1}^H A_{[\text{CLS}],i}^{(L,h)}. \quad (1)$$

This gives a simple summary of how strongly the sentence-level classification token attends to each input token.

Gradient $\times$ input. Let e_i denote the embedding of token i and $z_{\hat{y}}$ the logit of the model’s predicted class. We compute

$$s_i^{\text{g} \times \text{x}} = \left\langle e_i, \frac{\partial z_{\hat{y}}}{\partial e_i} \right\rangle. \quad (2)$$

We take the absolute value so that strongly supportive and strongly inhibitory tokens are both treated as important, following the magnitude-based use of gradient attributions [1].

Leave-one-out deletion. For each token i , we delete that token once, re-run the model, and measure the drop in predicted-class probability:

$$s_i^{\text{loo}} = p_{\hat{y}}(x) - p_{\hat{y}}(x_{\setminus i}), \quad (3)$$

where $x_{\setminus i}$ is the sequence after removing token i . This method directly estimates how much the current prediction depends on each token [5].

3.3. Evaluation Protocol

We evaluate the three rankings in two ways.

Ranking agreement. For each validation example, we compute pairwise Spearman correlation between the token rankings produced by the three methods. This measures whether different importance definitions rank tokens similarly.

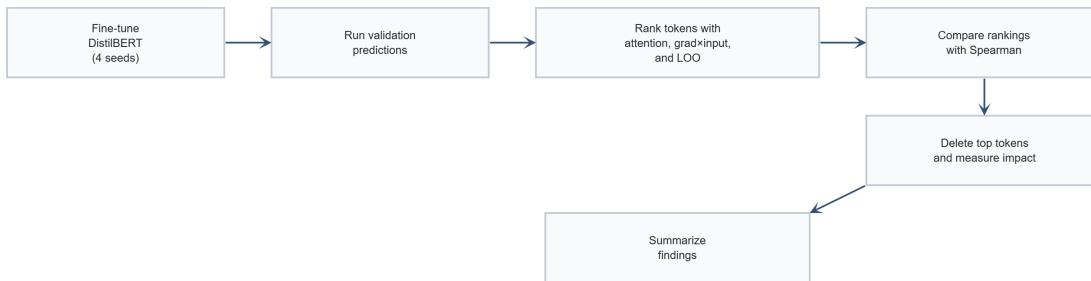


Figure 1: Overview of the experiment pipeline. For each seed, we fine-tune DistilBERT on SST-2, run validation predictions, compute three token-importance rankings, compare the rankings with Spearman correlation, and evaluate them through token deletion.

Deletion tests. For each method, we delete the top- k ranked tokens for $k \in \{1, 2, 3\}$ and compare the new prediction against the original one. We report two main metrics. The first is confidence drop,

$$\Delta c = \max_y p_y(x) - \max_y p_y(x_{\setminus S_k}), \quad (4)$$

where S_k is the selected token set. The second is prediction flip rate, i.e., the fraction of examples whose predicted label changes after deletion. We also keep a random deletion baseline by sampling tokens uniformly from the candidate token positions and averaging across 5 random trials per example. Our main paper focuses on fixed top- k deletion because it is the most interpretable and stable comparison across methods.

4. Results

4.1. Base Model Performance Is Stable Across Seeds

Before analyzing token importance, we first verify that the classifier itself behaves consistently across random seeds. Table 1 shows the validation metrics for the four runs. Performance is stable, with mean accuracy 0.9074 ± 0.0052 and mean F1 0.9094 ± 0.0044 . This reduces the risk that the interpretability conclusions are driven by a single outlier run.

4.2. Attention Has Weak Agreement with Alternative Rankings

Figure 2 summarizes pairwise rank correlation between token-importance methods. Attention agrees only weakly with the alternatives: its mean Spearman correlation is 0.224 ± 0.011 against gradient $\times$ input and 0.176 ± 0.001 against LOO. Agreement between gradient $\times$ input and LOO is even lower at 0.087 ± 0.006 .

Two conclusions follow from this result. First, raw attention does not behave like a close surrogate for the more

Seed	Accuracy	F1	Loss
42	0.9128	0.9152	0.3180
43	0.9048	0.9089	0.3474
44	0.9117	0.9108	0.2759
45	0.9002	0.9028	0.3766
Mean $\pm$ std	0.9074 ± 0.0052	0.9094 ± 0.0044	0.3295 ± 0.0372

Table 1: Validation performance on SST-2 across four random seeds.

Agreement between ranking methods

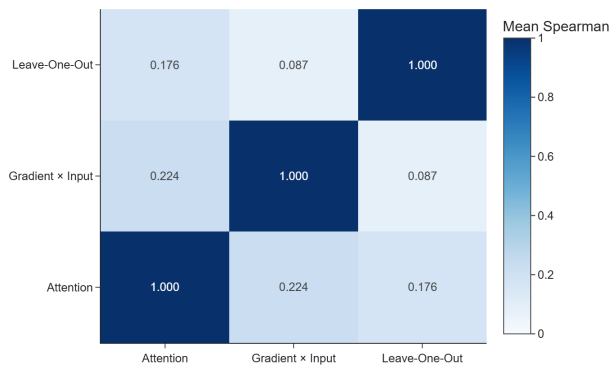


Figure 2: Mean Spearman rank correlation between token-ranking methods, aggregated across the four seeds. Agreement is weak, especially between attention and LOO.

intervention-oriented LOO measure. Second, token importance is itself definition-dependent: even two non-attention alternatives may highlight different aspects of the same pre-diction. Therefore, if attention is used as an explanation, it should not be assumed to match other attribution methods by default.

4.3. LOO Identifies the Most Disruptive Tokens

Agreement between rankings is only part of the story. A second question is whether a method identifies tokens whose deletion actually changes the model’s behavior. Figure 3 shows the top- k deletion results.

LOO is clearly the strongest method under both evaluation metrics. Deleting the single top LOO-ranked token causes mean confidence drop 0.0383 and flip rate 0.302, compared with 0.0138 and 0.139 for attention, 0.0073 and 0.081 for gradient $\times$ input, and 0.0030 and 0.038 for random deletion. The same ordering remains at $k = 2$ and $k = 3$: LOO remains strongest, attention remains above random, and attention also exceeds gradient $\times$ input in this specific setup.

This pattern is important because it separates two claims. Attention is not a reliable proxy for token importance in the strong sense of matching LOO or ranking tokens similarly to other methods. However, it is not useless either. Its deletion curves show that attention contains non-trivial information about influential tokens, just substantially less than the direct perturbation signal provided by LOO.

4.4. A Qualitative Case Shows Why Attention Can Mislead

Figure 4 illustrates a validation example on which the model predicts *positive* for a *negative* gold label. This case is useful because it exposes which tokens most support the model’s *actual* decision, even when that decision is wrong.

The sentence contains the word *compelling*, which LOO ranks first. Deleting it reduces the model’s positive confidence from 0.9983 to 0.5009, showing that it is the most influential word for the mistaken prediction. By contrast, raw attention assigns its highest score to the final period, and deleting that token leaves the prediction almost unchanged. Gradient $\times$ input highlights more content-bearing tokens than attention, but still identifies a much weaker token than LOO in this example. The case therefore matches the aggregate picture: attention can sometimes highlight superficially salient positions without capturing the token that most strongly drives the model output.

5. Discussion

Our results suggest a nuanced conclusion. On the one hand, raw last-layer attention is not a reliable standalone proxy for token importance. Its agreement with both gradient $\times$ input and LOO is weak, and its deletion behavior is consistently less disruptive than LOO. If one interprets token importance as “which token most changes the prediction when removed,” then attention is clearly not the best choice in this setting.

On the other hand, attention is not completely detached from model behavior. Attention-based deletion is stronger

than random deletion throughout the fixed top- k experiments, which means it captures some non-random signal about influential tokens. In our setup it also outperforms gradient $\times$ input on both confidence drop and flip rate. This matters because it prevents an overly strong conclusion. The empirical result is not that attention is meaningless; rather, it is that attention is incomplete and unreliable when used as a direct substitute for stronger importance measures.

One plausible explanation is that raw Transformer attention mixes several functions at once. Prior analyses of BERT attention show that heads often focus on delimiters, punctuation, or positional patterns in addition to semantically meaningful content [2, 9]. Averaging last-layer attention from [CLS] may therefore preserve some useful task signal while also inheriting structural attention behaviors that do not correspond closely to causal influence on the output.

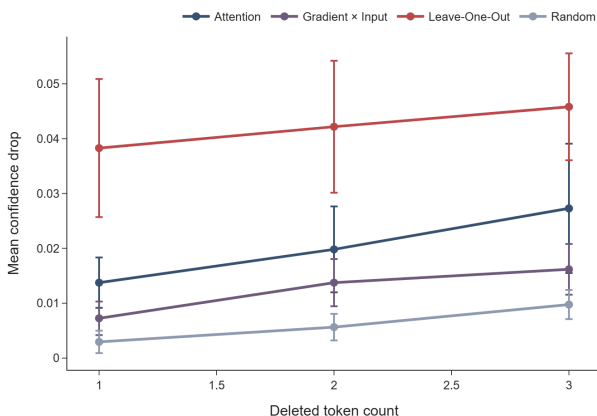
The project also has several limitations. First, it uses only one task and one model family, so the conclusions should not be generalized automatically to other datasets or larger language models. Second, the analysis is performed at the wordpiece level, which can split meaningful words into multiple ranked tokens. Third, we examine only one attention summary: mean last-layer [CLS] attention. Different layers, individual heads, or effective attention variants may behave differently. Finally, LOO itself is a local perturbation method rather than a complete causal account, and it is computationally more expensive than attention or gradient-based scoring.

6. Conclusion

We revisited a simple but important interpretability question: do attention weights reflect token importance in a Transformer sentiment classifier? In a controlled DistilBERT-on-SST-2 setting with four random seeds, the answer is largely no. Attention shows only weak agreement with gradient $\times$ input and LOO rankings, and LOO-selected tokens are much more effective at reducing confidence and flipping predictions when deleted.

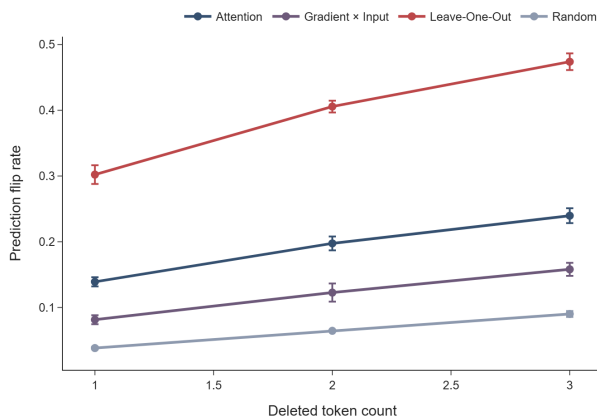
At the same time, attention is not purely noise. Its deletion results are consistently stronger than random deletion, indicating that it carries some useful information about the model’s behavior. The most accurate summary of our findings is therefore that attention can be informative, but it should not be treated as a faithful or sufficient explanation on its own. Future work could test the same question on more tasks and models, compare additional attribution methods, and analyze whether specific layers or heads provide more reliable signals than the simple last-layer attention summary used here.

Mean confidence drop under fixed top-k deletion



(a) Mean confidence drop.

Prediction flip rate under fixed top-k deletion



(b) Prediction flip rate.

Figure 3: Top- k deletion results aggregated across four seeds. Error bars denote standard deviation across seeds. LOO-selected tokens produce the largest disruption to the classifier, while attention-based deletion is consistently stronger than random deletion and stronger than gradient $\times$ input deletion in this setting.

Validation example 519 token rankings

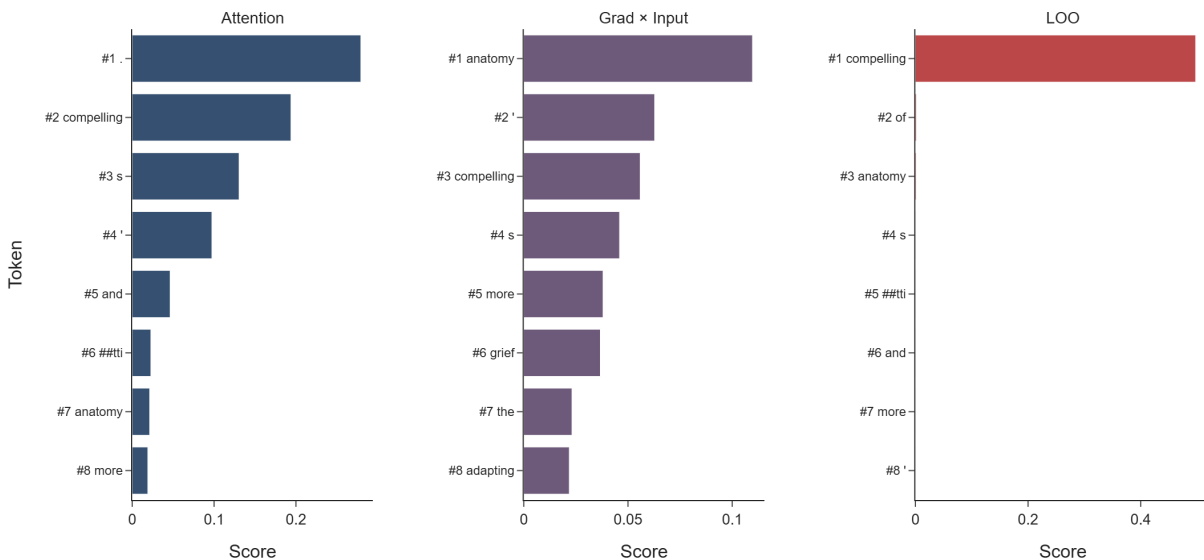


Figure 4: Qualitative example 519 from seed 42. The model predicts *positive* with confidence 0.9983 although the gold label is *negative*. Attention peaks on punctuation, whereas LOO identifies *compelling* as the main driver of the error.

References

- [1] M. Ancona, E. Ceolini, C. Oztireli, and M. Gross. Towards better understanding of gradient-based attribution methods for deep neural networks. In *International Conference on Learning Representations*, 2018. 2
- [2] K. Clark, U. Khandelwal, O. Levy, and C. D. Manning. What does BERT look at? an analysis of BERT’s attention. In *Proceedings of the 2019 ACL Workshop BlackboxNLP: Analyzing and Interpreting Neural Networks for NLP*, pages 276–286. Association for Computational Linguistics, 2019. 2, 4
- [3] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186. Association

for Computational Linguistics, 2019. 1, 2

[4] S. Jain and B. C. Wallace. Attention is not explanation. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 3543–3556. Association for Computational Linguistics, 2019. 1, 2

[5] J. Li, W. Monroe, and D. Jurafsky. Understanding neural networks through representation erasure. *arXiv preprint arXiv:1612.08220*, 2017. 2

[6] V. Sanh, L. Debut, J. Chaumond, and T. Wolf. DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*, 2019. 1, 2

[7] S. Serrano and N. A. Smith. Is attention interpretable? In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 2931–2951. Association for Computational Linguistics, 2019. 1, 2

[8] R. Socher, A. Perelygin, J. Wu, J. Chuang, C. D. Manning, A. Ng, and C. Potts. Recursive deep models for semantic compositionality over a sentiment treebank. In *Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing*, pages 1631–1642. Association for Computational Linguistics, 2013. 1, 2

[9] K. Sun and A. Marasovic. Effective attention sheds light on interpretability. In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, pages 4126–4135. Association for Computational Linguistics, 2021. 2, 4

[10] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin. Attention is all you need. *arXiv preprint arXiv:1706.03762*, 2017. 1

[11] A. Wang, A. Singh, J. Michael, F. Hill, O. Levy, and S. R. Bowman. GLUE: A multi-task benchmark and analysis platform for natural language understanding. In *Proceedings of the 2018 EMNLP Workshop BlackboxNLP: Analyzing and Interpreting Neural Networks for NLP*, pages 353–355. Association for Computational Linguistics, 2018. 1, 2

[12] S. Wiegreffe and Y. Pinter. Attention is not not explanation. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 11–20. Association for Computational Linguistics, 2019. 1, 2